

Credit Risk & Loan Default Prediction

Explainable Machine Learning + SQL + Power BI

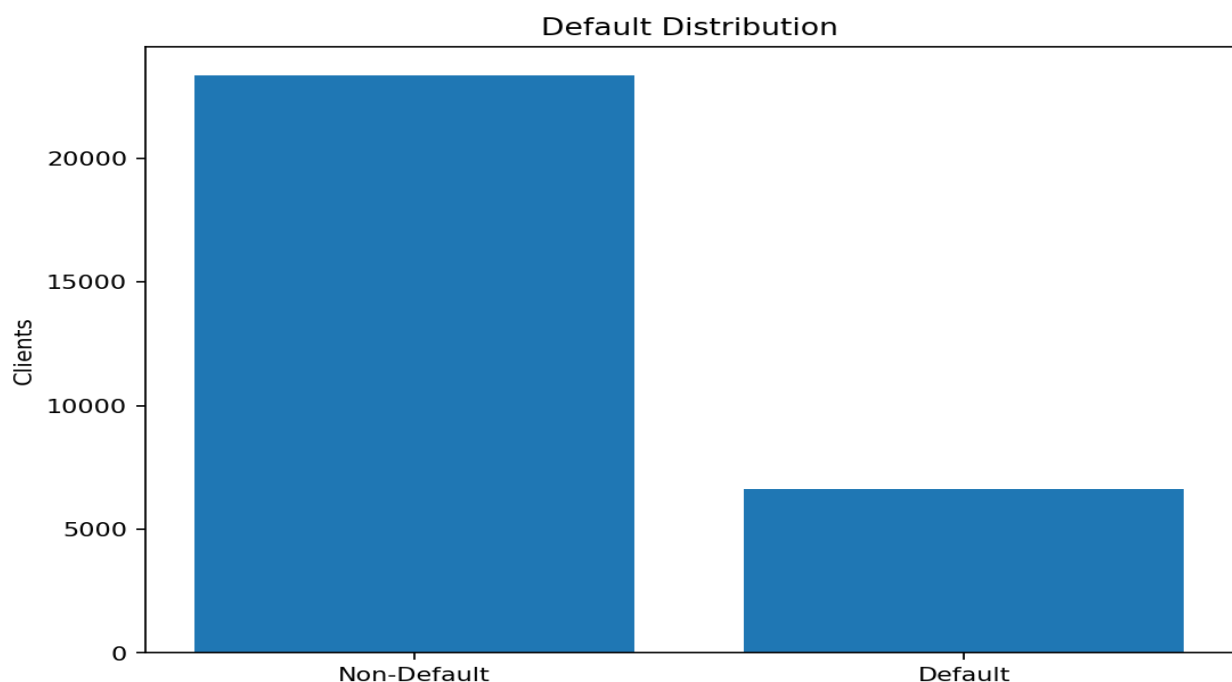
Portfolio analytics project using the UCI Default of Credit Card Clients dataset

Executive Summary

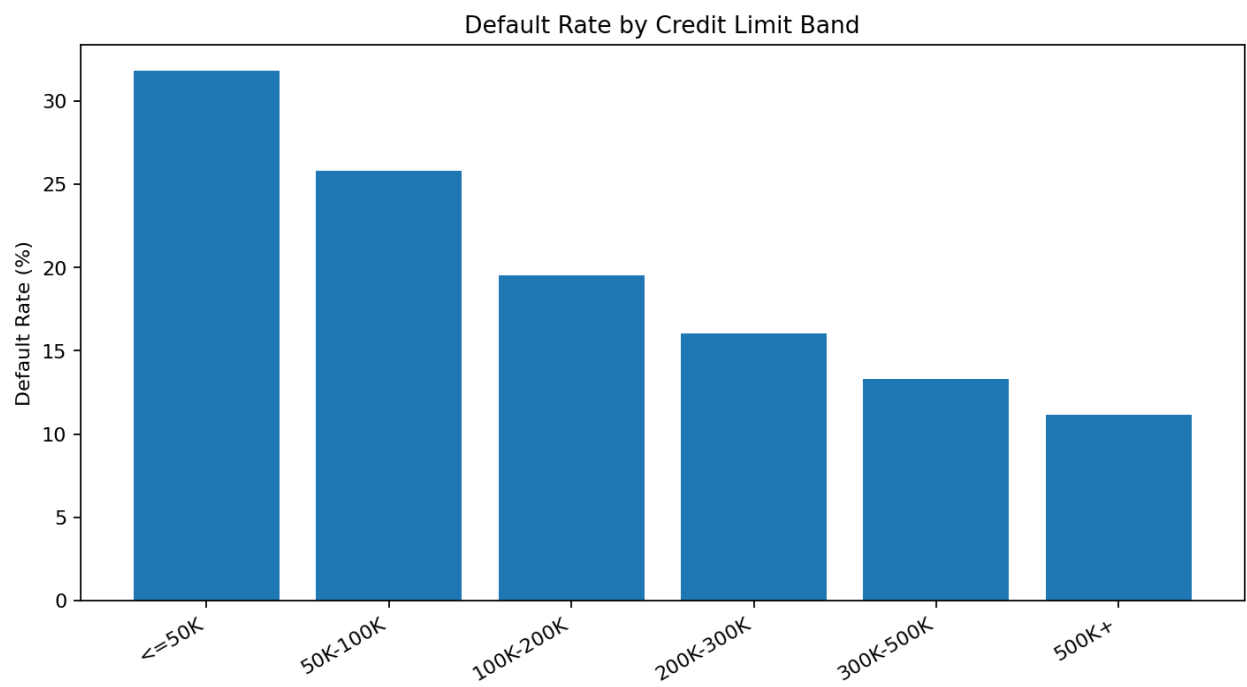
The cleaned analytical dataset contains 29,965 client records, including 6,630 observed defaults, giving an overall default rate of 22.13%. Four classification models were compared using imbalance-aware metrics. Random Forest achieved the highest PR-AUC (0.560) and was selected for portfolio risk scoring. At the standard 0.50 threshold it achieved ROC-AUC 0.779, recall 0.593, precision 0.494, and F1 0.539.

The probability-based segmentation placed 8,113 clients in the High-risk band. The observed default rate in that band was 54.55%, compared with 12.73% for Medium risk and 2.05% for Low risk. The High-risk band represented a credit-limit exposure proxy of 939,663,680.

Metric	Actual Result
Clients analysed	29,965
Observed defaults	6,630
Overall default rate	22.13%
Selected model	Random Forest
Selected model ROC-AUC	0.779
Selected model PR-AUC	0.560
Selected model Recall	0.593
Best test F1 threshold	0.50
High-risk clients	8,113
High-risk observed default rate	54.55%



Observed default distribution

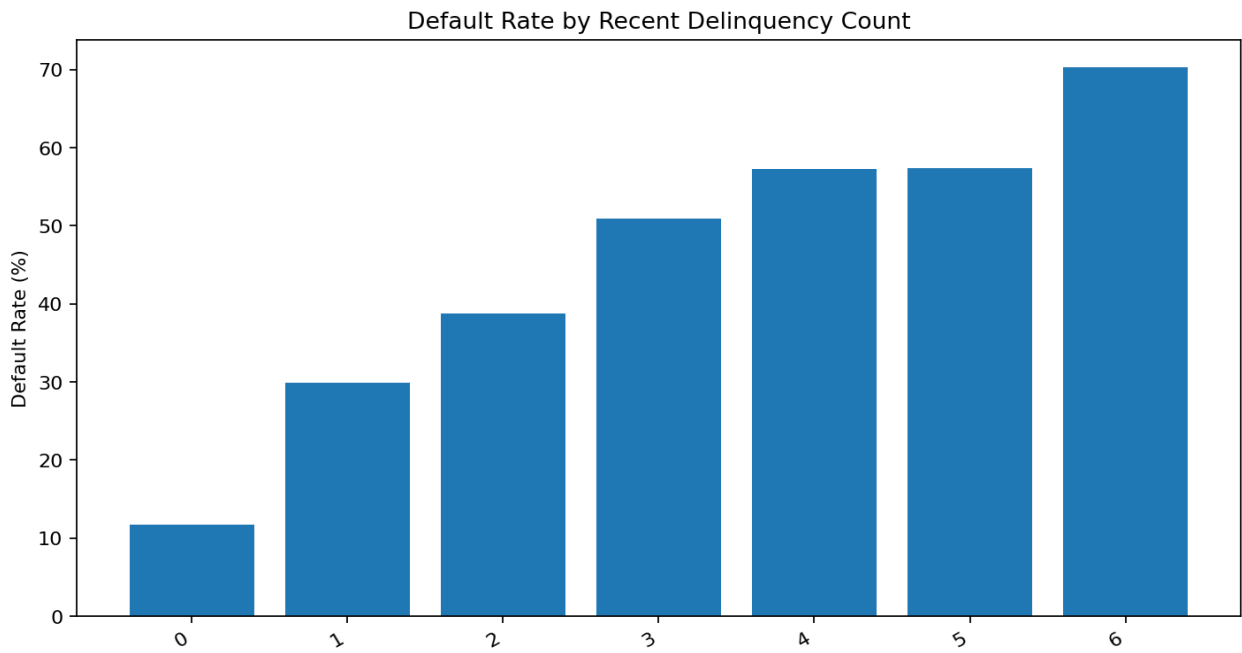


Observed default rate by credit-limit band

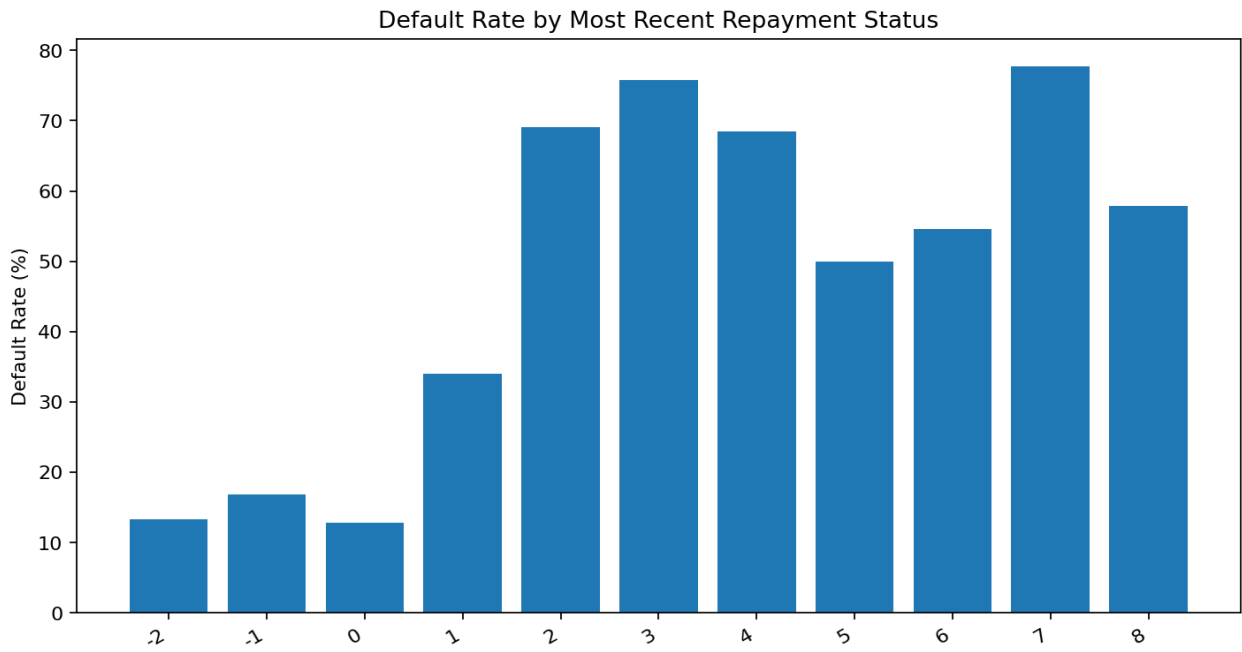
Credit Risk Patterns

The descriptive analysis shows a strong relationship between repayment behaviour and observed default. Clients with 5-6 delinquent months had an observed default rate of 67.95%, while clients with no delinquent months had a rate of 11.71%. Lower payment-to-bill ratios were also associated with higher observed default rates in this dataset.

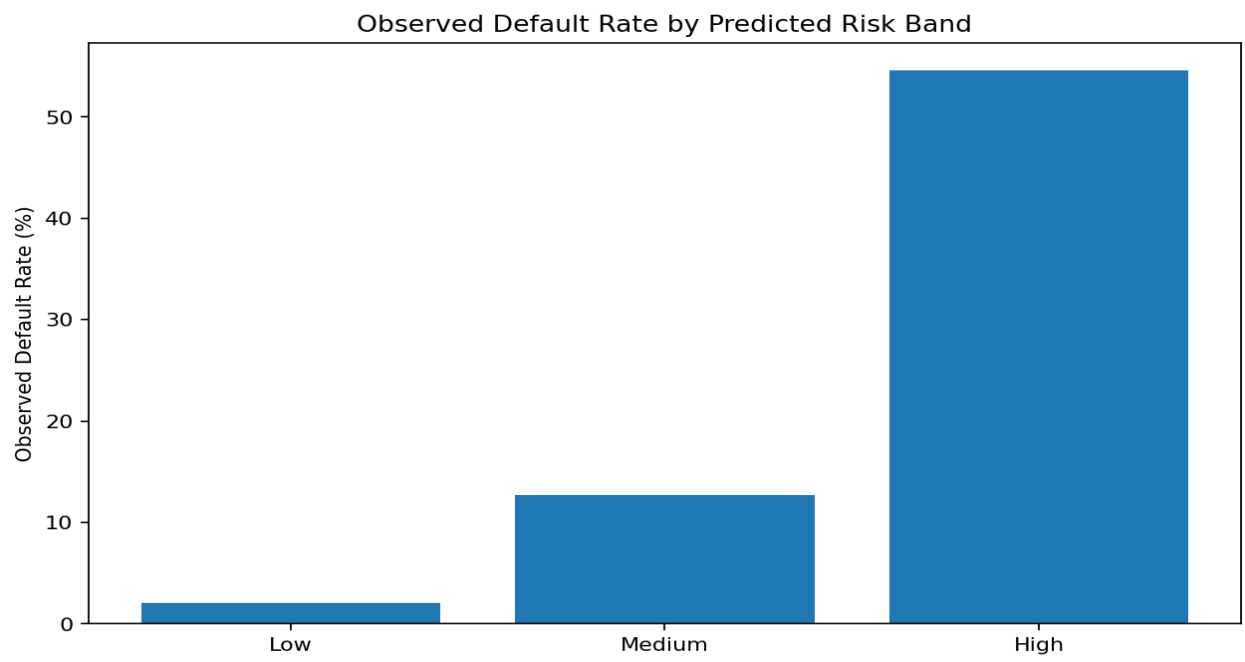
Segment	Clients	Observed Default Rate	Credit Limit Exposure
Low	5,406	2.05%	1,468,560,000
Medium	16,446	12.73%	2,609,176,000
High	8,113	54.55%	939,663,680



Default rate rises with repeated delinquency



Observed default rate by most recent repayment status

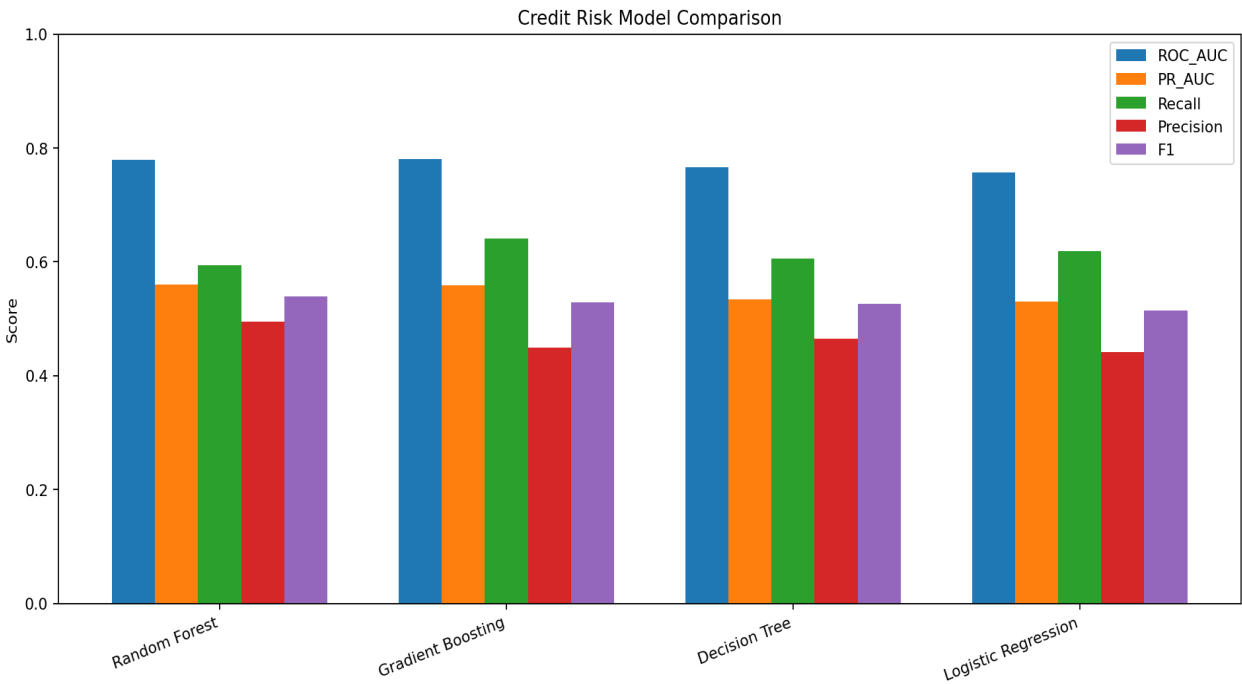


Observed default rate by predicted risk band

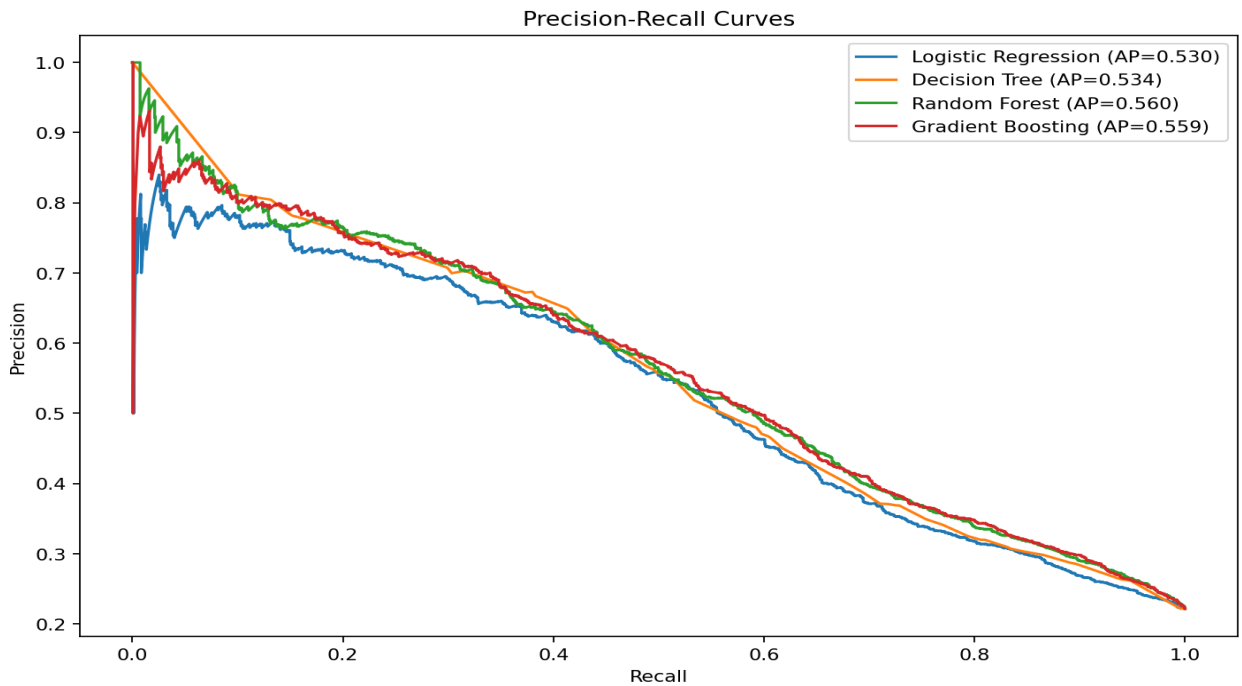
Model Evaluation

Because default is the minority class, the model comparison uses PR-AUC, ROC-AUC, Recall, Precision and F1 rather than relying on Accuracy alone. Random Forest had the highest PR-AUC in the test set. Gradient Boosting produced a slightly higher ROC-AUC, but its PR-AUC was lower. The model-selection rule therefore favoured Random Forest.

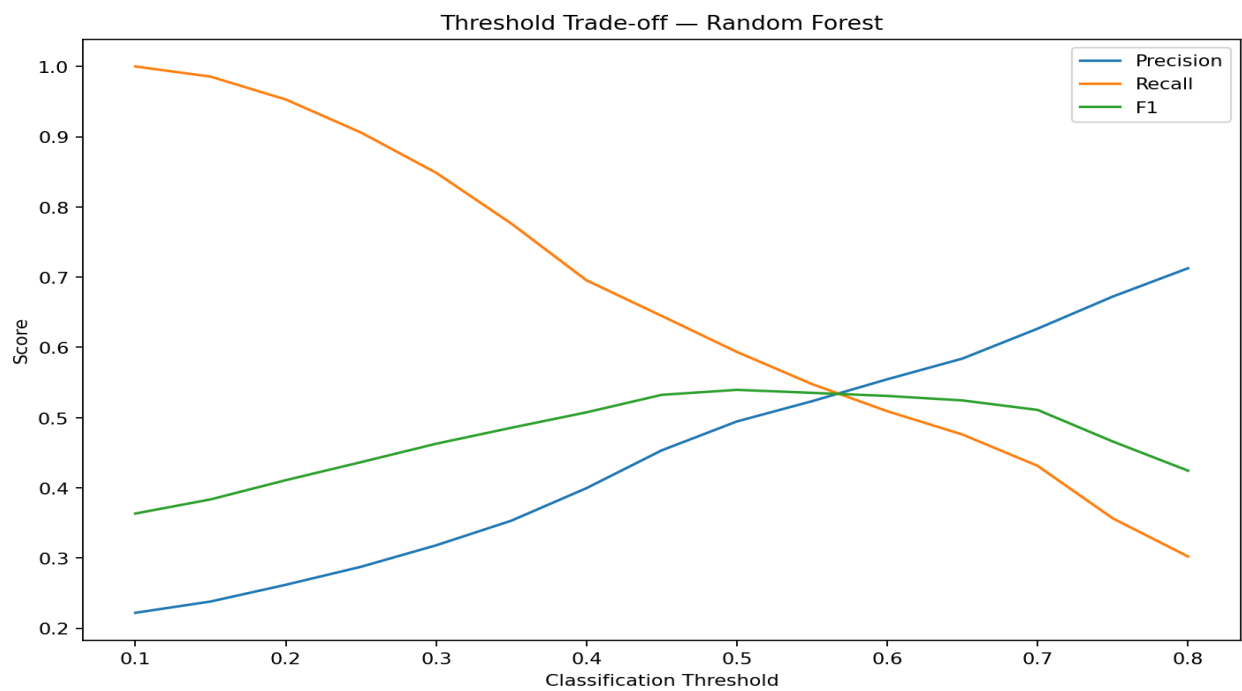
Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
Random Forest	0.776	0.494	0.593	0.539	0.779	0.560
Gradient Boosting	0.747	0.449	0.641	0.528	0.781	0.559
Decision Tree	0.759	0.465	0.606	0.526	0.767	0.534
Logistic Regression	0.742	0.441	0.619	0.515	0.758	0.530



Model comparison across key classification metrics



Precision-recall curves

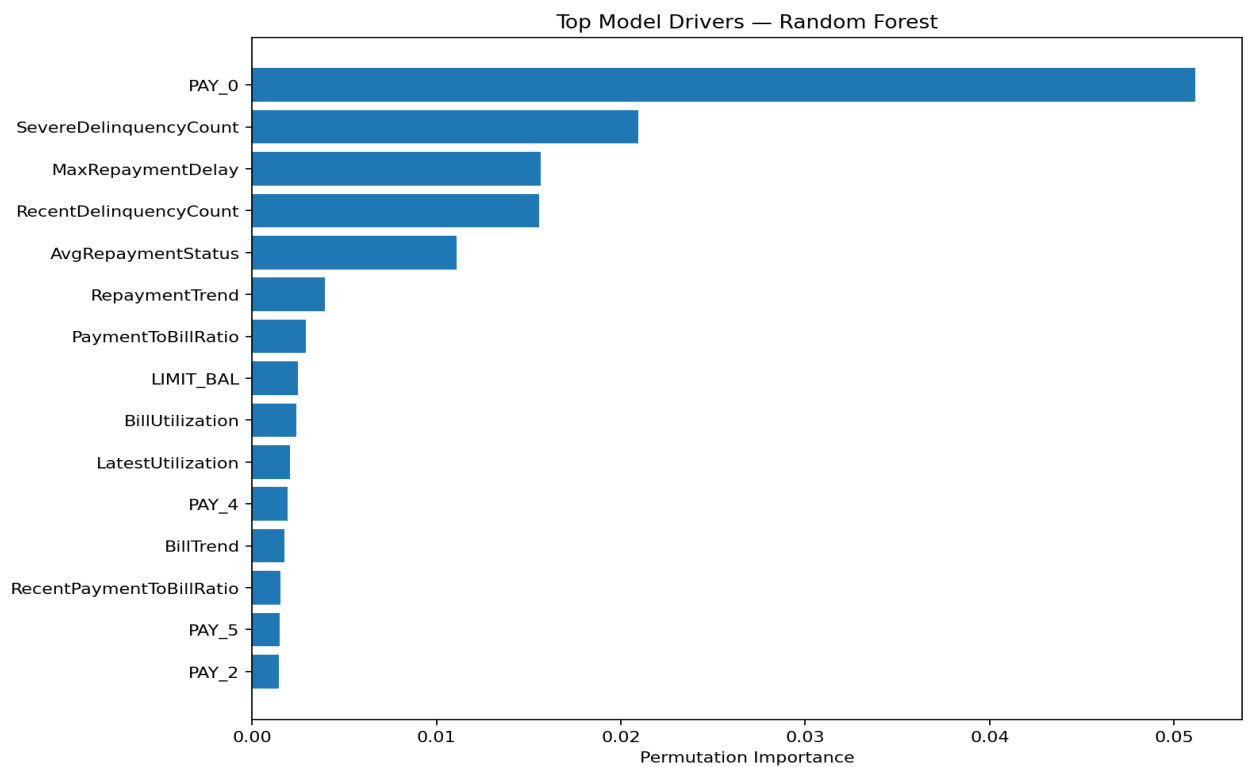


Precision, recall and F1 across probability thresholds

Explainability & Responsible Use

Permutation importance indicates that recent repayment behaviour is the strongest source of predictive signal in the selected model. PAY_0, severe delinquency count, maximum repayment delay, recent delinquency count and average repayment status are the leading features. Feature importance describes how the model behaves; it does not prove that a variable causes default.

Rank	Feature	Permutation Importance
1	PAY_0	0.0511
2	SevereDelinquencyCount	0.0210
3	MaxRepaymentDelay	0.0156
4	RecentDelinquencyCount	0.0156
5	AvgRepaymentStatus	0.0111
6	RepaymentTrend	0.0040
7	PaymentToBillRatio	0.0029
8	LIMIT_BAL	0.0025
9	BillUtilization	0.0024
10	LatestUtilization	0.0021



Top global model drivers from permutation importance

Responsible Modelling Decisions

Sensitive demographic variables (sex, education, marital status and age) were excluded from the predictive feature set and retained only for descriptive audit views. This reduces one obvious pathway for demographic attributes to directly drive portfolio risk scores, but it does not by itself establish fairness because other variables can still act as proxies.

A real lending deployment would require calibration, formal fairness testing, regulatory and policy review, monitoring for drift, documented threshold selection, adverse-action explainability where applicable, and human oversight. This project is educational portfolio analytics only and is not an automated lending approval or rejection system.

Key Portfolio Conclusions

1. Recent repayment behaviour and repeated delinquency provide the strongest predictive signal in this dataset.
2. The High-risk probability band has a materially higher observed default rate than the Medium and Low bands.
3. Threshold selection creates a real trade-off between capturing more defaults and increasing false positives.
4. PR-AUC is particularly useful here because the target is imbalanced and the positive class is the business risk event.